

SDLC2 for AI-assisted delivery

Governance · traceability · human authority ·
operational learning

AI compresses execution time. It does not resolve ambiguity.

THE ACCELERATION

Agents can translate intent into plans, code, tests and diagnostics at a speed that changes the economics of iteration.

THE CONTROL PROBLEM

The same speed magnifies unclear authority, scope drift, hidden assumptions, accidental cross-component changes and premature claims of 'done'.

SDLC2 is controlled transformation – not ceremony.

AUTHORITY

Humans own intent, approval, accepted risk and release. Agents never self-approve.

SCOPE

Use cases, architecture and component-scoped tasks bound the change.

EVIDENCE

Observable proof determines progress. Build, review, deployment and operation remain separate states.

Twelve lifecycle areas form one learning loop.

UNDERSTAND · 01–03

Context & Domain
Intent
System Model & Use Cases

PROVE & DESIGN · 04–06

Benchmarks & Success
Viable Proof of Concept
Architecture

CONTROL CHANGE · 07–09

Planning
Specifications as Code
Build & Test

REVIEW & LEARN · 10–12

Review & Release
Operational Reality
Insights & Learning → 01

Documents are decision interfaces between humans and agents.

CONTROLLED INPUTS

Intent defines why. Use cases define behavior. Architecture defines ownership and trust boundaries. Sprint plans define scope, evidence and exclusions.

EXECUTABLE CONSTRAINTS

Contracts, schemas, invariants, acceptance examples and tests make specifications agent-ready. Documents govern work; they do not silently authorize production or risk.

Approval is bound to the exact revision – not the idea.

01 · AGENT DRAFT

Visible revision

The file exists unstaged, reviewer initials are blank, and the provided link opens the exact proposal.

02 · HUMAN REVIEW

Identity + exact bytes

A registered human enters initials and stages the complete reviewed contents. Partial staging is not approval.

03 · EXECUTION GATE

Proceed while valid

Coding proceeds only while governed documents remain approved. Any edit invalidates approval and resets the gate.

AGENTS.md is the repository's durable operating contract.

MANDATORY CONTEXT

Read governance before plans, docs, code, tests or releases.

NON-NEGOTIABLES

Exact approval, ≤ 50 -line authored files, human-staged deletions, no force-push.

AUTONOMOUS ZONE

Inside approved scope: implement, build, test, diagnose and perform reversible checks.

EXPLICIT CONSENT

Ask before scope change, production risk, secrets, money, security or irreversible action.

Granular approval makes bounded agent autonomy possible.

01 · SPRINT PLAN

Authorize the increment

Goal, in-scope behavior, exclusions, success evidence and decision gates are reviewed first.

02 · PLAN TASKS

Constrain the edits

A separate approved document orders the work. Every task names exactly one component and its tests.

03 · BUILD + VALIDATE

Evidence

Execute in approved scope. Cross-component checks are read-only; fixes require a new task and approval.

A green build is not a release—and a release is not reality.

DEFINITION

Planned
Human-approved

CONSTRUCTION

Implemented
Tested

DELIVERY

Reviewed · committed · merged
Deployed · released

REALITY

Observed with real users, data, cost and consequences
Learning returned to Step 01

AI Shop evolves by resolving one uncertainty at a time.

SPRINT 001

Secure camera-to-AI proof

Native iPhone capture, server-side OpenAI credential and a physical-device camera-to-result loop.

SPRINT 002

Depth and breadth

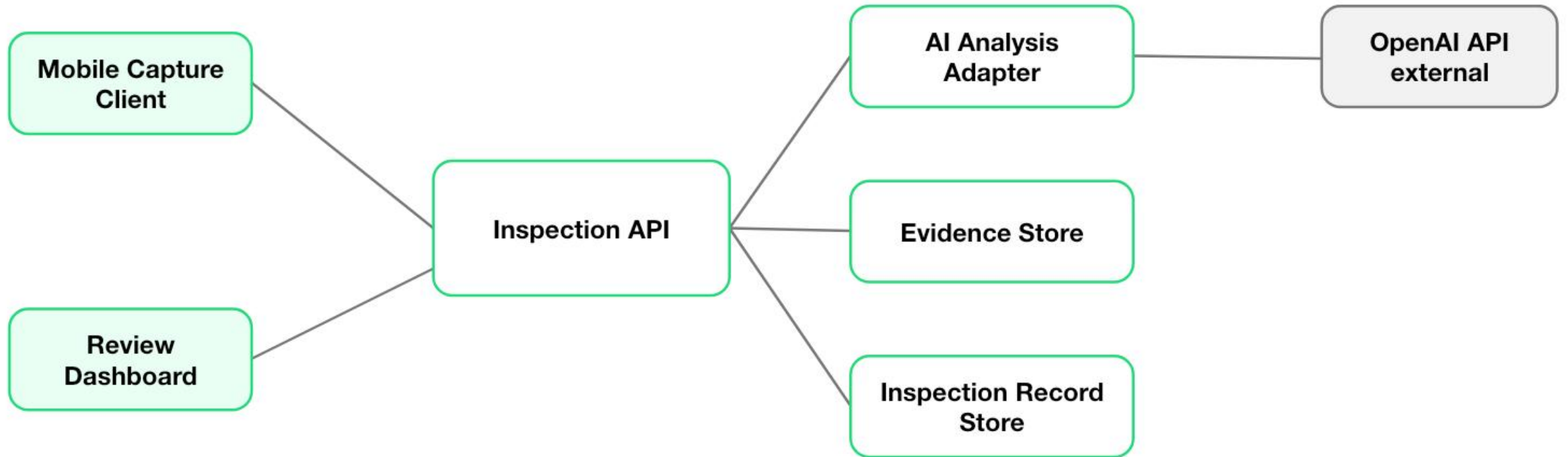
Target Product and Area Scan create mode-specific structured reports while communicating uncertainty.

SPRINT 003

Auditable inspection

Exact JPEG, SHA-256, immutable AI findings, private storage and attributable review.

Governed component boundaries become architecture.



Invariant: original evidence and initial AI findings remain immutable; human review is separate and attributable.

Fourteen tasks. Six owned components. One ordered system.

CONTRACT & MODEL · T1-2

Inspection API submission contract
AI Adapter target-position behavior

CAPTURE & EVIDENCE · T3-4, 6, 13

Original iPhone JPEG bytes
Hash, private storage, retrieval and bucket

RECORDS & API · T5, 7-9, 11-12, 14

Immutable records, reviewer operations, failure semantics, Firestore
and JPEG compatibility

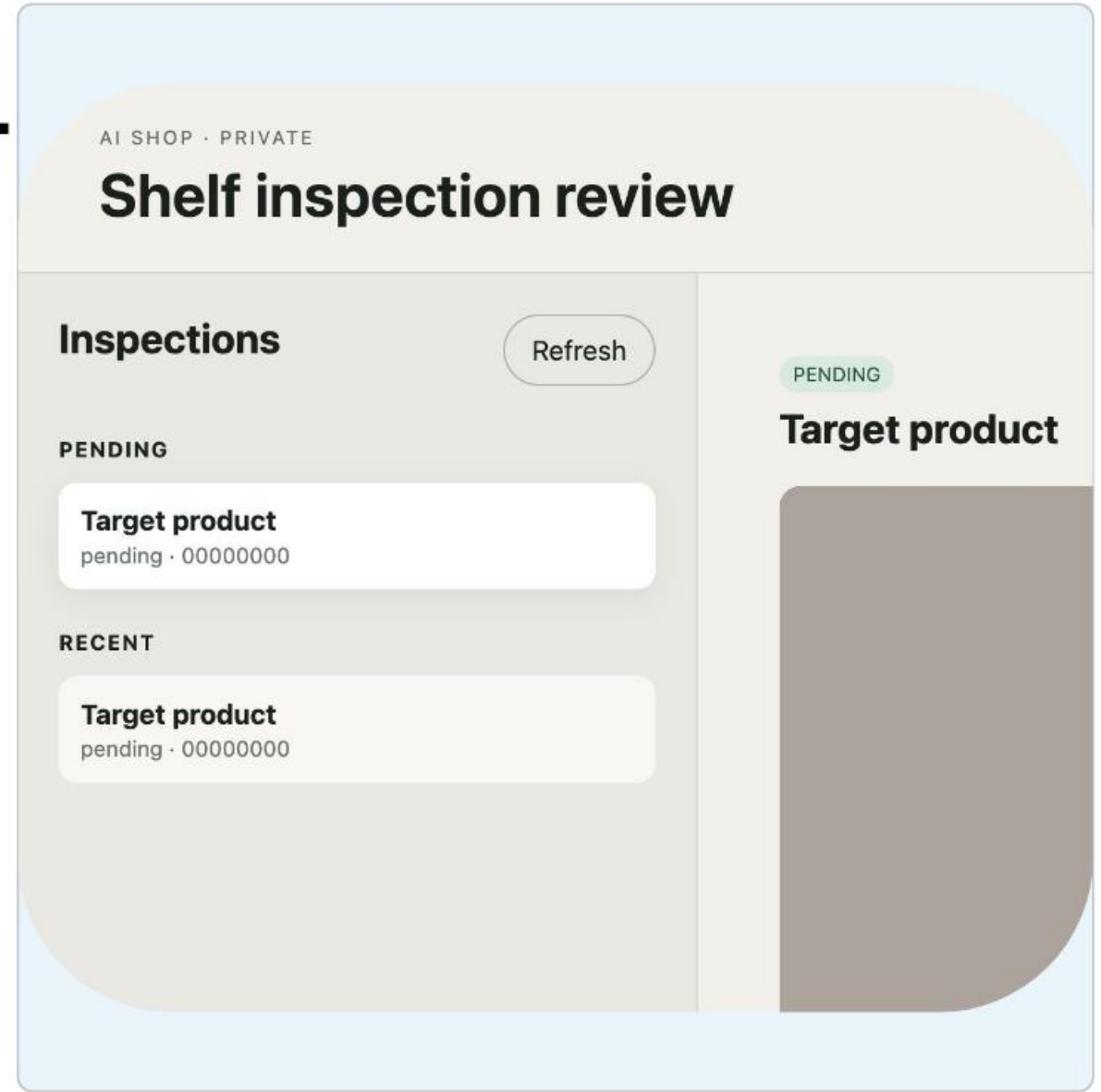
REVIEW & VALIDATION · T10 + GATE

Private dashboard
Read-only cross-component validation
Any fix becomes a new task

The audit record preserves both evidence and disagreement.

AN EVIDENCE MODEL, NOT JUST A UI

Each scan connects the exact original JPEG and SHA-256 with immutable initial findings. An authorized reviewer records Verified, Corrected or Rejected separately—with identity, note and server time. Unauthorized access and distinct failure modes are tested.



The method reduces risk by making friction explicit.

WHAT IT PREVENTS

Scope drift, agent self-approval, stale approvals, partial staging, cross-component blast radius, invisible assumptions and status inflation.

WHAT IT COSTS

More small documents, disciplined staging, slower starts and a real human bottleneck. The rigor should be proportionate—but consequential agent-assisted change benefits from it.

Inspect the method in the repository

github.com/pabloelustondo/aishop

Intent · use cases · architecture · sprint
controls

Native iOS · Firebase API · evidence store
· review dashboard